

Database Management - Notes

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Session 1: Data Management

What is a Database

A database is an organized and structured set of records stored and managed for a common purpose. A data record is an information element which, as a unit, describes a complex set of facts.

Architecture and Components of an Information System

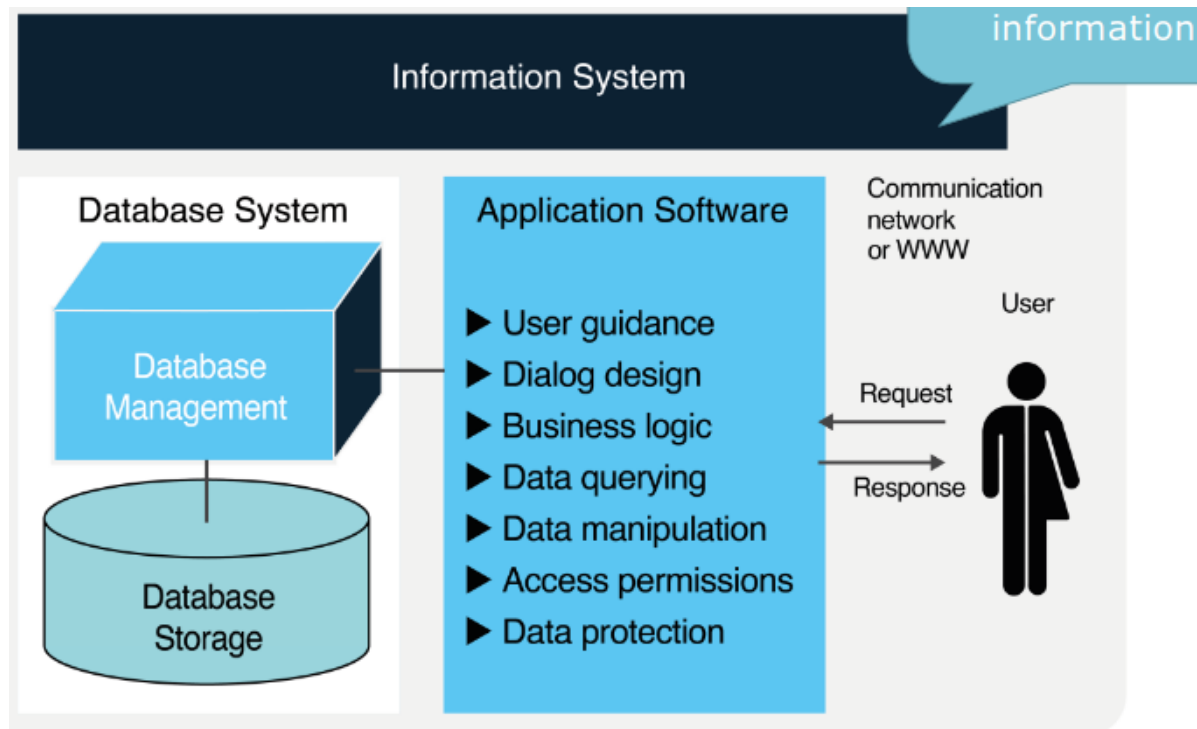


Figure 1: Information System

File System

A file system is a structured method used by an operating system to organize, store, and retrieve data on a storage drive. It manages how data is named, placed, and grouped into files and directories, acting as a bridge between the raw hardware and the user.

Why do we use databases?

- Enable multi-user operations: Manage transactions
- Query and manipulate data sets
- Ensure data consistency and integrity automatically
- Retrieve large data quantities (indexing, parallelization) efficiently

Big Data Challenges

Big data primarily refers to data sets that are too large or complex to be dealt with by traditional data-processing software. Data with many entries (rows) offers greater statistical power, while data with higher complexity (more attributes or columns) may lead to a higher false discovery rate.

- **Volume:** This refers to the sheer amount of data generated every second, requiring massive storage capacities that go far beyond traditional hardware.
- **Velocity:** This represents the incredible speed at which data is created, processed, and analyzed, often in real-time or near real-time.
- **Variety:** This describes the diverse types of data available, ranging from structured text in databases to unstructured formats like video, images, and social media posts.
- **Veracity:** This focuses on the quality, accuracy, and trustworthiness of the data, as “messy” data can lead to incorrect conclusions.
- **Value:** This is the most critical V, referring to the ability to turn bulk data into meaningful insights that drive business decisions or solve problems.

NoSQL Databases

Optimization for Big Data:

- Non-relational data structures
- Non-relational database languages

Data Management

Data Management comprises all methods related to creating value from data as a resource.

Data management: Creating value with data

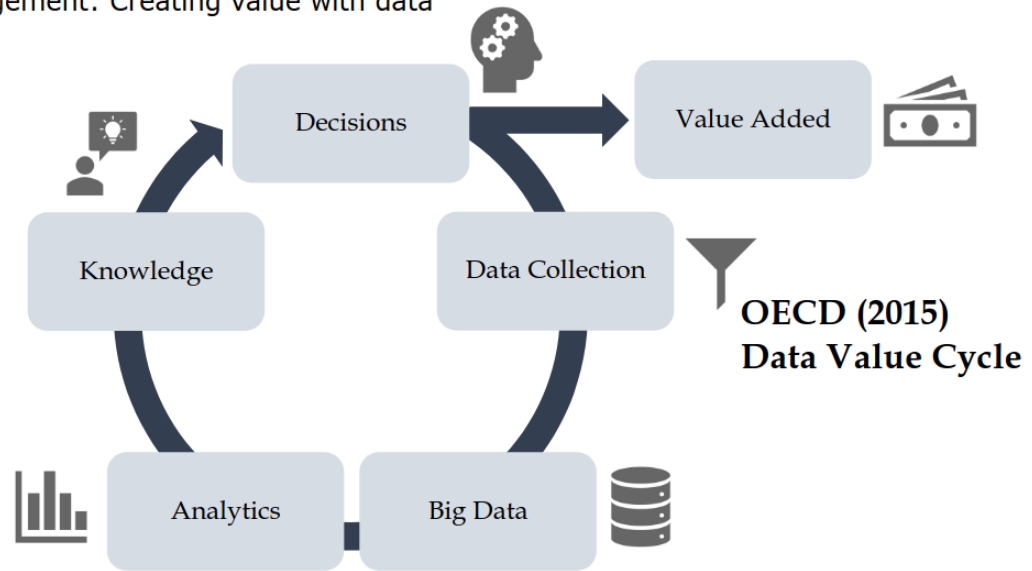


Figure 2: Data Value Cycle

Database Application

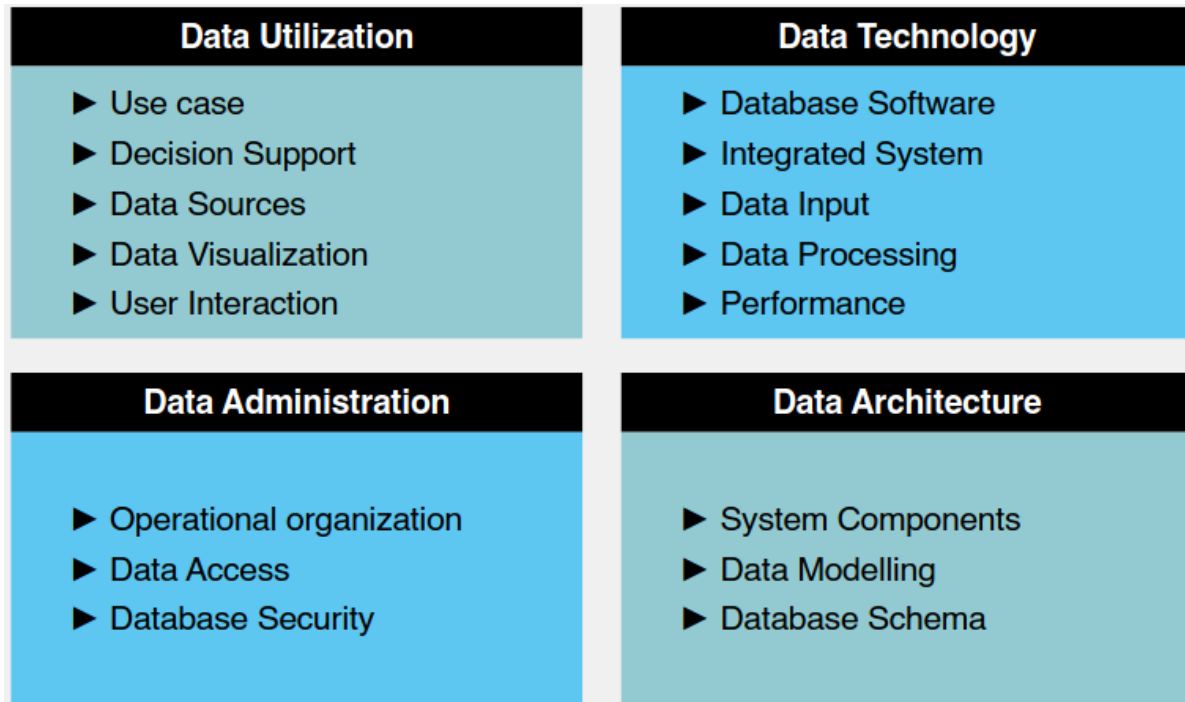


Figure 3: Database Applications

Session 2: Database Modeling

Database Modeling

Requirements Analysis

1. Requirements analysis		Order no. 113
		Date: 07/14/2023
Goal	For project monitoring purposes, employees, work, and project times should periodically be reported per department.	
1.	Employees report to departments, with each employee being assigned to exactly one department.	
2.	Each project is centrally assigned a unique project number.	
3.	Employees can work on multiple projects simultaneously; the respective percentages of their time are logged.	
4.	...	

Figure 4: Requirement Analysis

Conceptual Modeling

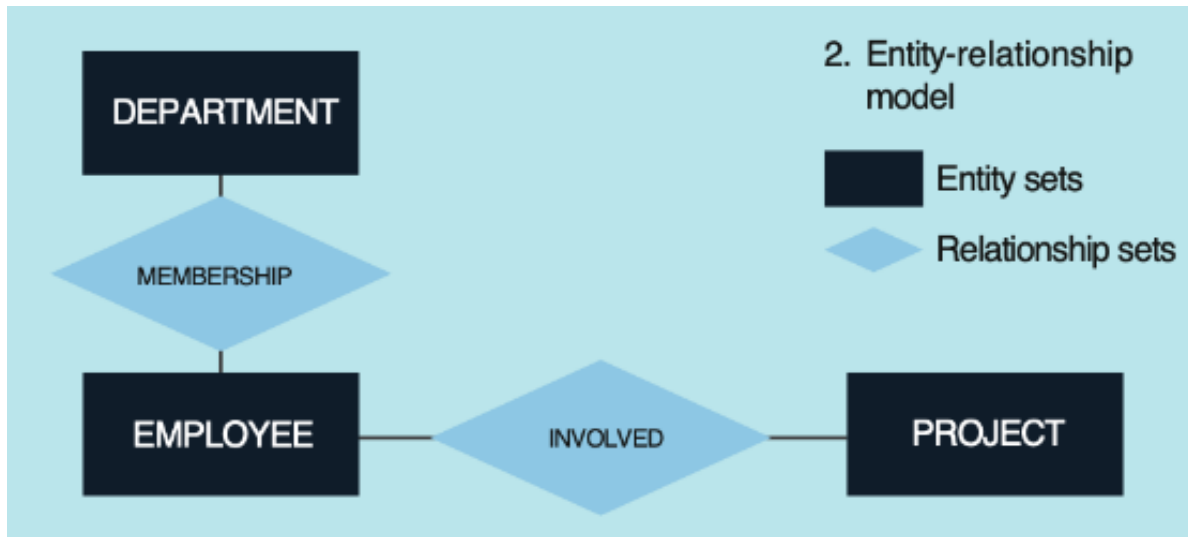


Figure 5: Conceptual Modeling

Database Shema

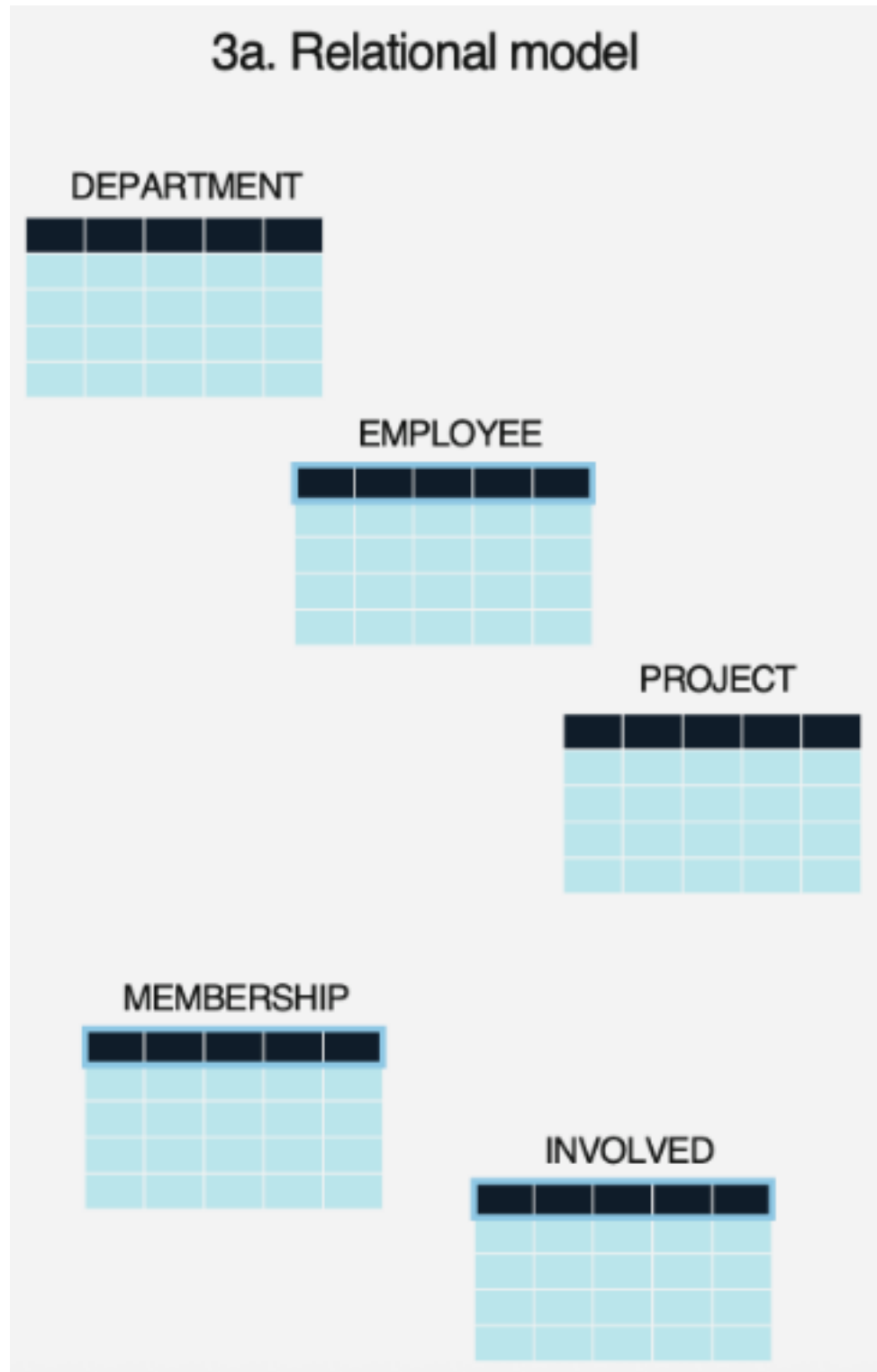


Figure 6: Database Shema

Table Defintion

A table or relation refers to a set of tuples (set of ordered, immutable objects) shown as a table.

- Table name: A table has a unambiguous name.
- Attribute name: Is unambiguous; denotes a specific column with the desired attribute.
- No column order: The attribute number is unlimited; the column order has no meaning.
- No row order: The tuple number in a table is arbitrary; the tuple order has no meaning.
- Identification key: One of the attributes or combination thereof unambiguously identifies the tuples in the table and is called the identification key

Entities

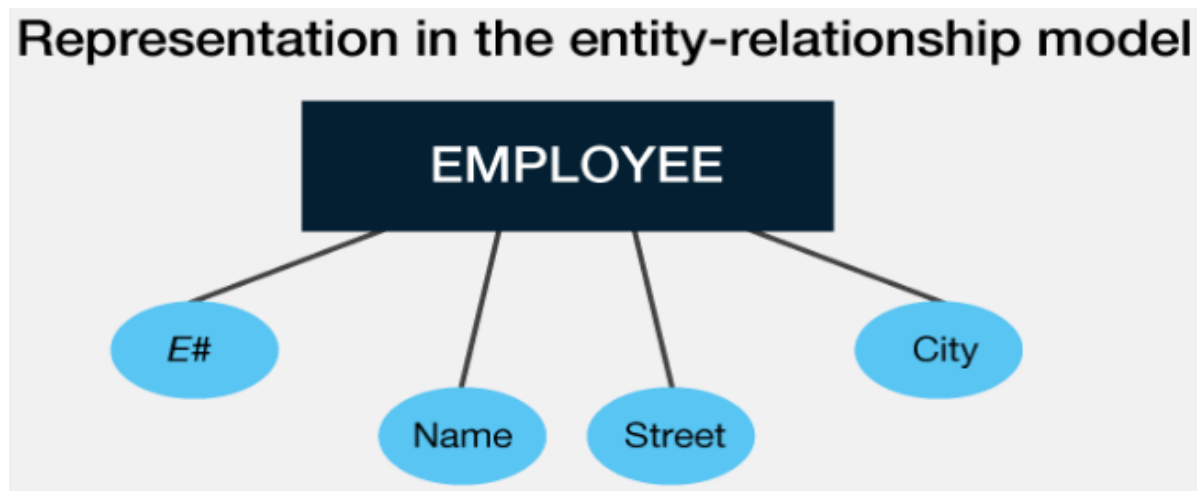


Figure 7: Table Entity

Relationships

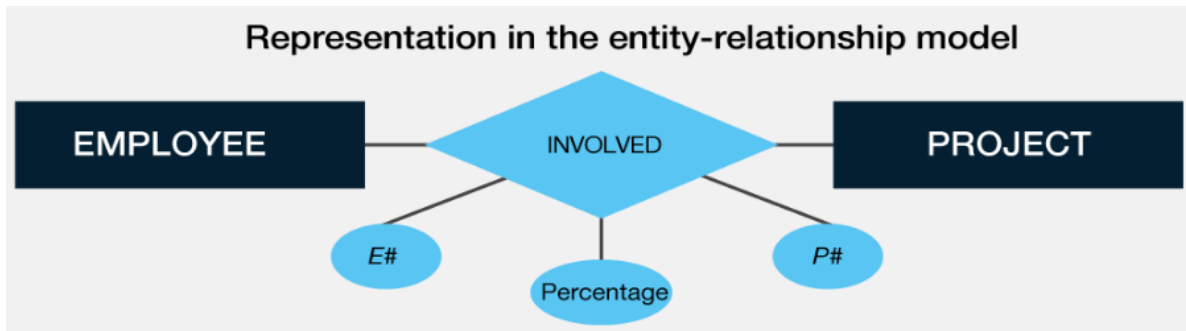


Figure 8: Entity Relationship

The Entity Relationship Model

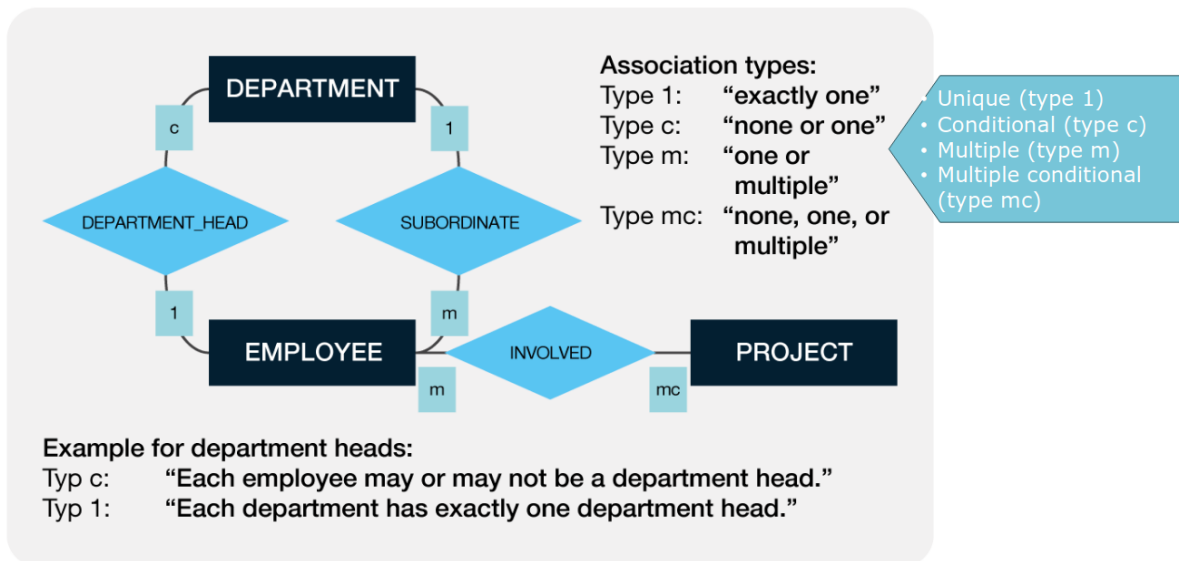


Figure 9: The Entity Relationship Model

Redundancy and Normal Forms in Tables

Redundancy in tables can lead to anomalies in databases. An attribute is redundant if individual values of it can be omitted without any loss of information. Normal forms serve to avoid redundancy.

Note: Anomalies = Insertion anomaly, mutation anomaly, deletion anomaly.

Normal Forms

Stage	Key Focus	Goal
1NF	Atomicity	Eliminate duplicate columns and multi-valued cells.
2NF	Full Dependency	Eliminate partial dependencies on the primary key.
3NF	Direct Dependency	Eliminate dependencies between non-key columns.
4NF	Multi-valued facts	Isolate independent multi-valued relationships.
5NF	Reconstruction	Minimize redundancy in complex many-to-many relationships.

Session 3: Database Integration

SQL Data Manipulation Lanugage

- CREATE TABLE: Define tables
- DROP TABLE: Delete tables
- ALTER TABLE: Change table definition
- INSERT: Insert data

```
INSERT INTO Employee VALUES  
( 'E22', 'Cole', '7th Street', 'Manhattan', 'A6'),  
( 'E23', 'Smith', 'Reseda Blvd', 'Los Angeles', 'A5')
```

- UPDATE: Changing the data
- SELECT: Filter data
- DELETE: Delete data

Data Base Integration

Database Integration not only helps automate Data Preparation. Metadata and schema integration also contributes to data understanding.

ETL (Extract, Transform, Load)

In the traditional ETL process, data is transformed on a secondary staging server before it ever reaches the target database (usually a Data Warehouse).

Note: Best for structured data, small-to-medium data volumes, and legacy on-premise systems.

```
LOAD DATA LOCAL INFILE '/path/to/ratings.csv'  
INTO TABLE Ratings  
FIELDS TERMINATED BY ','  
ENCLOSED BY ''''  
IGNORE 1 LINES  
(UserID, MovieID, Rating, @Timestamp)  
SET Timestamp = FROM_UNIXTIME(@Timestamp);
```

Loading large Datasets

By Table Import Utility in the MySQL Shell.

ELT (Extract, Load, Transform)

ELT is a modern evolution where data is loaded into the target system (usually a Cloud Data Warehouse or Data Lake) in its raw form. The transformation happens within the target system using its own processing power.

Note: Best for unstructured/Big Data, high velocity, and modern cloud environments (like Snowflake, BigQuery, or Redshift).

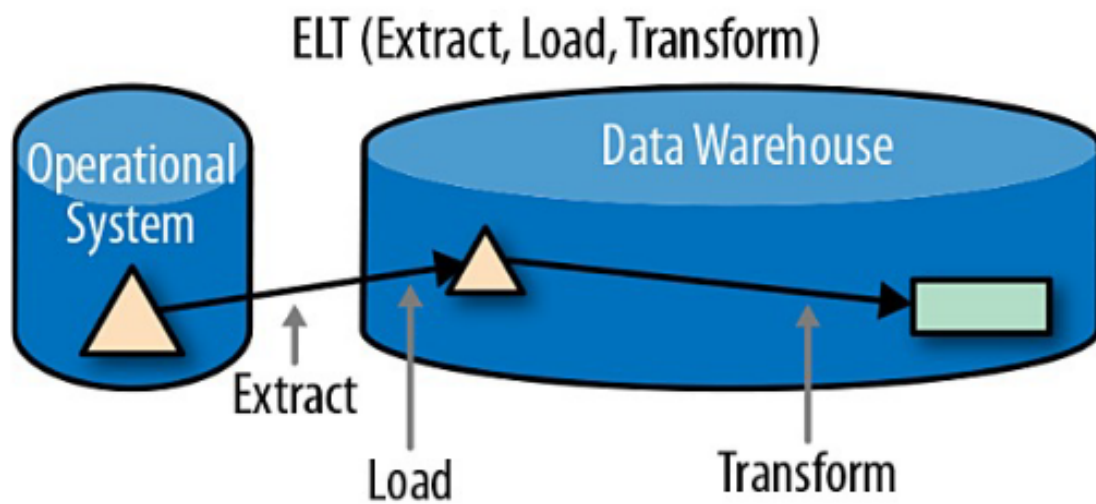
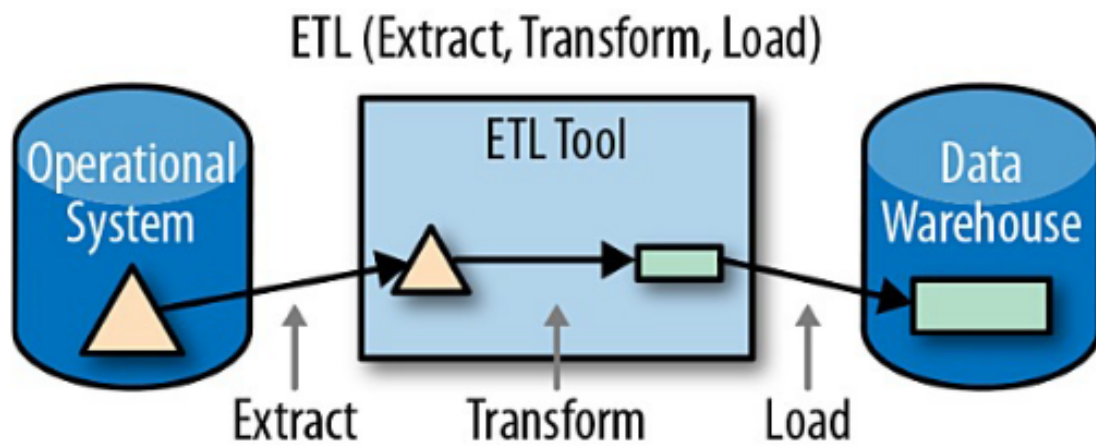


Figure 10: ETL vs. ELT

Session 5: SQL Language Concepts

Data Definition Language (DDL)

Tables

```
CREATE TABLE person (  
    PersonId INTEGER,  
    Name VARCHAR(100),  
    Birthdate DATE);
```



Tip

Always use semicolon to terminate SQL statements

Data Types

Data Type	Category	Description	Typical Use Case
INT	Numeric	A standard integer (4 bytes). Range is -2,147,483,648 to 2,147,483,647 (or 0 to 4,294,967,295 if UNSIGNED).	Primary keys, user IDs, quantities.
BIGINT	Numeric	A large integer (8 bytes). Used when values might exceed the INT range.	High-traffic IDs, auto-incrementing logs, financial transactions.
DECIMAL(p,s)	Numeric	Fixed-point number where p is total precision (digits) and s is scale (decimals). Exact, no rounding errors.	Currency, exact measurements, financial data (e.g., DECIMAL(10,2)).
FLOAT / DOUBLE	Numeric	Floating-point numbers for approximate values. DOUBLE has higher precision than FLOAT.	Scientific calculations, GPS coordinates where slight rounding is okay.

Data Type	Category	Description	Typical Use Case
VARCHAR(M)	String	Variable-length string. M defines the maximum character length (up to 65,535). Efficient for varying lengths.	Names, email addresses, passwords, titles.
CHAR(M)	String	Fixed-length string. Right-padded with spaces to specified length M (up to 255). Faster than VARCHAR for fixed data.	Country codes (US, CA), MD5 hashes, state abbreviations.
TEXT	String	Long-form text data (up to 65,535 characters). Cannot have a default value.	Blog post bodies, product descriptions, comments.
DATE	Date/Time	Date value in YYYY-MM-DD format. Range: 1001-01-01 to 9999-12-31.	Birthdays, hiring dates, expiration dates.
DATETIME	Date/Time	Date and time combination in YYYY-MM-DD HH:MM:SS format. Supported range is 1000-01-01 to 9999-12-31.	Event schedules, booking times.
TIMESTAMP	Date/Time	Date and time combination, stored as UTC seconds since epoch. Range: 1970-01-01 to 2038-01-19. Automatically updates.	“Created at” or “Updated at” record tracking.
BOOLEAN	Numeric	Synonym for TINYINT(1) . Zero is considered false, non-zero values are true.	Flags like <code>is_active</code> , <code>has_paid</code> , <code>is_admin</code> .

Data Type	Category	Description	Typical Use Case
JSON	Document	Stores RFC-compliant JSON documents. Enables native schema-less data validation and quick lookups.	User preferences, dynamic form data, third-party API responses.

Note: MySQL data types.

Primary-/Foreign-Keys

```
CREATE TABLE course (
  CourseId INTEGER NOT NULL,
  CourseNo VARCHAR(20),
  ProfessorPersonId INTEGER,
  CONSTRAINT PK_Course
  PRIMARY KEY (CourseId),
  CONSTRAINT FK_Course_Person
  FOREIGN KEY (ProfessorPersonId)
  REFERENCES Person (PersonId)
);
```

Surrogate keys are often numbers. DBMS supports you to automatically generate these numbers by using `AUTO_INCREMENT` (as Primary Key). This command automatically creates a numeric value.

Constraints

- Unique: Cannot insert duplicates in a column
- Check: Ensures that the values in the column meet the given criteria
- Default: Inserts the default value if column is not part of the INSERT statement
- Null: You use NULL columns when values are not always provided
- Not Null: Use NOT NULL for columns if values must be provided

Altering Tables

```
ALTER TABLE person
ADD City VARCHAR(50) NOT NULL;

ALTER TABLE person
MODIFY City SET VARCHAR(100) NULL;

ALTER TABLE person
DROP COLUMN City;

ALTER TABLE person
ADD CONSTRAINT CK_Person_Birthdate
CHECK (YEAR(Birthdate) >= 1900);
```

Temporary Tables

```
CREATE TEMPORARY TABLE temp_mature_person AS
SELECT
    p.PersonId, p.Name, p.Birthdate
FROM
    person AS p
WHERE
    p.Birthdate < '1960-01-01';
```

Note

You may use WHERE 1 = 0 to create an empty (temporary) table.

Note: Use CREATE TABLE to copy a table

Views

Views doesn't actually store any data of its own. Instead, it stores a predefined **SELECT** query, and whenever you query the view, MySQL runs that underlying query behind the scenes and presents the results just like a regular table.

```

CREATE VIEW registrated_students AS
SELECT
    c.CourseId, c.CourseNo,
    sc.StudentPersonId,
    sc.RegistrationDate
FROM
    studentsincourse AS sc
INNER JOIN
    course AS c
ON
    sc.CourseId = c.CourseId
WHERE
    sc.CancellationDate IS NULL;

```

Data Query Language (DQL)

Grouping / Aggregation

Group sets by one or more columns and calculate aggregations. Grouping columns must be part of the **GROUP BY** clause. Functions on grouped columns must also be applied in the **GROUP BY** and **SELECT** clause.

```

SELECT
    c.ProfessorPersonId,
    LEFT(c.SemesterNo, 4) AS CourseYear,
    SUM(c.LessonsPerWeek) AS WeeklyQuota
FROM
    course AS c
WHERE
    c.LessonsPerWeek < 4
GROUP BY
    c.ProfessorPersonId,
    LEFT(c.SemesterNo, 4);

```

Grouping with Selection

You can filter aggregated values with the **HAVING** clause.

1. Filter the dataset with **WHERE** before grouping
2. Aggregate the values with **GROUP BY**
3. Filter the aggregated values with **HAVING** after grouping

```

SELECT
    c.ProfessorPersonId,
    LEFT(c.SemesterNo, 4) AS CourseYear,
    SUM(c.LessonsPerWeek) AS WeeklyQuota
FROM
    course AS c
WHERE
    c.LessonsPerWeek < 4
GROUP BY
    c.ProfessorPersonId,
    LEFT(c.SemesterNo, 4)
HAVING
    SUM(c.LessonsPerWeek) > 4;

```

Derived Table

Often used to pre-calculate values in the inner query and reuse these values several times in the outer query.

```

SELECT
    p.PersonId,
    p.Name,
    s.Registrations
FROM
    person AS p
INNER JOIN (
    SELECT
        sc.StudentPersonId AS Id,
        COUNT(*) AS Registrations
    FROM
        studentsincourse AS sc
    GROUP BY
        sc.StudentPersonId
) AS s
ON p.PersonId = s.Id;

```

Correlated Subquery

Correlated subqueries reference one or more column(s) in the outer query.

```

SELECT
    e.CourseId,
    e.ExamNote,
    e.StudentPersonId
FROM
    examresult AS e
WHERE
    e.ExamNote > (
        SELECT
            AVG(r.ExamNote)
        FROM
            examresult AS r
        WHERE
            r.CourseId = e.CourseId
    );

```

Self-contained subquery

Self-contained subqueries are independent of the outer query.

```

SELECT
    e.StudentPersonId,
    e.ExamNote
FROM examresult AS e
WHERE e.ExamNote > (
    SELECT
        AVG(n.ExamNote)
    FROM
        examresult AS n
);

```

EXISTS / NOT EXISTS

Use a correlated subquery to check for (non-)existing rows in the correlated table. EXISTS has some advantages over IN / NOT IN predicate:

- You can use more than one predicate
- More consistent handling of NULL

```

SELECT
    sc.CourseId,
    sc.StudentPersonId,
    sc.RegistrationDate
FROM
    studentsincourse AS sc
WHERE
    sc.RegistrationDate >= '2019-04-01'
AND NOT EXISTS(
    SELECT
        *
    FROM
        examresult AS r
    WHERE
        sc.CourseId = r.CourseId
    AND
        sc.StudentPersonId = r.StudentPersonId
);

```

Cross Join

Creates a Cartesian product of the two involved tables.

```

SELECT
    p.PersonId, c.CourseId
FROM
    person AS p
CROSS JOIN
    Course AS c;

```

Outer Join

Add a table to the main table. Rows on the main table are preserved. Non-matching rows on the unpreserved side contain NULL.

```

SELECT
    p.PersonId,
    p.Name,
    p.Birthdate,
    c.CourseNo,

```

```

        c.SemesterNo
FROM
    person AS p
LEFT OUTER JOIN
    course AS c
ON
    p.PersonId = c.ProfessorPersonId
WHERE
    p.Birthdate < '1960-01-01';

```

Common Table Expressions

A Common Table Expression (CTE) is a temporary, named result set that you can reference within a larger statement.

```

WITH
    cteMatureProf AS (
        SELECT
            p.PersonId AS Id,
            p.Name,
            p.Birthdate
        FROM
            person AS p
        WHERE
            p.Birthdate < '1960-01-01'
    )

SELECT
    c.CourseNo,
    cM.Name,
    cM.Birthdate
FROM
    course AS c
LEFT OUTER JOIN
    cteMatureProf AS cM
ON
    c.ProfessorPersonId = cM.Id;

```

Note

The CTE is only valid within the current query.

CASE

Evaluates a list of expressions and returns the first one that is true.

```
SELECT
    c.CourseId,
    c.CourseNo,
CASE
    c.LessonsPerWeek
    WHEN 2 THEN 'Small'
    WHEN 3 THEN 'Middle'
    WHEN 4 THEN 'Large'
    ELSE 'Unknown'
END AS CourseSize
FROM
    course AS c;
```

Note: If you do not provide an ELSE clause, then NULL is returned when no match is found.

Window Function

Apply a function to a number of rows. The result will contain the detail rows (in contrast to group aggregations).

```
SELECT
    ROW_NUMBER() OVER(ORDER BY p.Name) AS RowNumber,
    p.Name,
    p.PersonId
FROM
    person AS p
WHERE
    p.PersonId < 10000
ORDER BY
    p.Name;
```

Partition Windows

You can split the set into several window frames. A common way is to partition the rows.


```

SELECT
    ROW_NUMBER() OVER (
        PARTITION BY c.ProfessorPersonId
        ORDER BY c.CourseId) AS RowNumber,
    c.ProfessorPersonId,
    c.CourseId
FROM
    course AS c
WHERE
    c.ProfessorPersonId < 2130;

```

Session 6: SQL Performance

Hardware Components for Data

Capacity and Latency

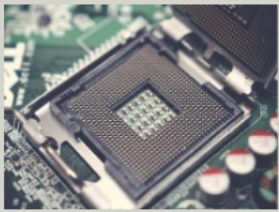
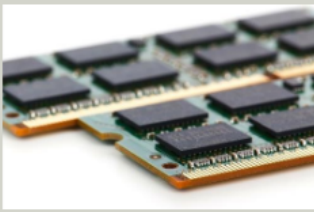
	CPU Cache	RAM/Memory	Disk (HDD/SSD)
			
Capacity	1MB – 50MB	Up to 128/256GB (more is possible)	Up to x TB
Access Latency	1 - 30ns	100 - 1'000ns	1 - 10ms (SSD: 0.05-0.15ms)
Scaled Latency*	2 sec - 1 min	4 - 40 min	SSD : 17 hrs - 4 days HDD: 1 - 9 months
*Baseline: • 1 CPU Cycle • Scaled to 1 sec 3-40x slower 1'000x !			

Figure 11: Hardware components for data

Bottlenecks

Table joins can be bottlenecks. Especially without indexes, they need to process a lot of data and, thus, might heavily use I/O. For query performance, you can skip/avoid joins by using:

- denormalized table design
- materialized view

Indexes

Do not just index each and every column in your tables (they must be updated with each insert, update and delete!). Create indexes that serve several different queries! When you load a huge amount of data, you might drop the indexes and recreate them after the import.

Cluster Indexes

- Tables physically ordered by key
- The key(s) and the data of a row are stored together
- Only one clustered index per table

Note: MySQL automatically clusters the table!

Nonclustered / Secondary Index

- Additional data structure containing key(s)
- More than one index possible per table
- Data is separate from index

```
CREATE INDEX  
    IDX_Person_Name  
ON  
    person (Name);
```

Heap

- Table content unordered
- Data usually appended to the end
- Identifier necessary to access the row (additional 6 bytes per row!)
 - Row Identifier [RID] or
 - Tuple Identifier [TID]

! Important

Heap is not an index.

Query Optimizer

The Query Optimizer supports you to reduce the amount of data processed and also saved in the interim result sets.

Algebraic Transformation

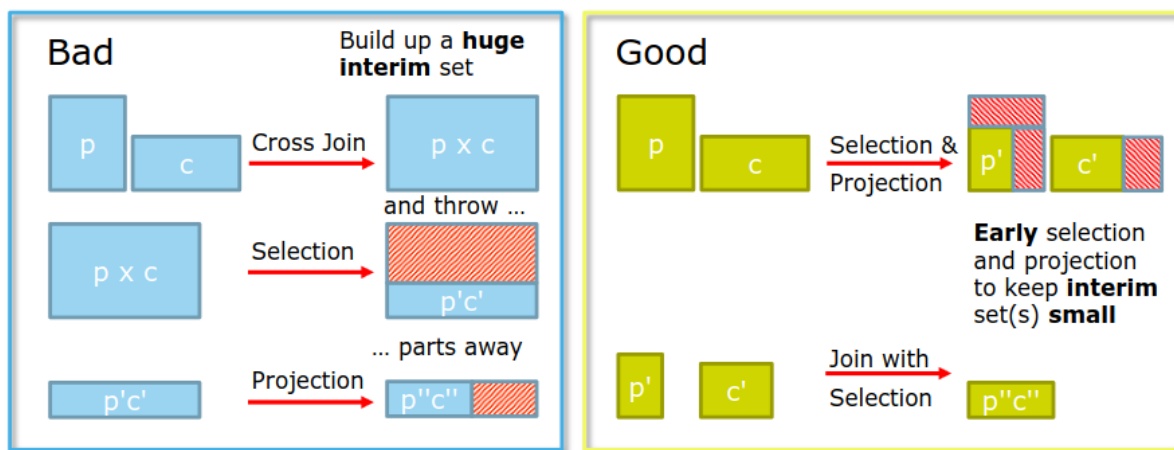


Figure 12: Relational algebra with(out) transformation

```
SELECT
    c.CourseNo,
    p.Name,
    p.Birthdate
FROM
    course AS c
INNER JOIN
    person AS p
ON
    c.ProfessorPersonId = p.PersonId
WHERE
    p.Name LIKE 'K%';
```

Execution Plan

Most DBMS provide a textual or graphical execution plan to estimate query performance.

The “Good” Operators (Efficient)

- Unique Key Lookup: Searching or joining via a Primary Key or Unique Index. (Fastest).
- Non-Unique Key Lookup: Searching an indexed column that allows duplicate values.
- Index Range Scan: Fetching a subset of data using operators like $\geq$, $<$, or BETWEEN ... AND.

The “Bad” Operators (Inefficient)

- Full Table Scan (“ALL”): MySQL reads every single row in the table from disk.
- Full Index Scan (“INDEX”): MySQL reads the entire index tree from top to bottom.

Syntactical Query Optimization

Functions on search columns

- Bad: `WHERE LEFT(b.Name, 2) = 'An';`
- Good: `WHERE b.Name LIKE 'An%';`

SARGable queries

SARGable stands for Search Argument Able. A query is SARGable if the MySQL engine can take advantage of an existing index to speed up the execution. If a query is Non-SARGable, MySQL cannot use the index and is forced to resort to a slow Full Table Scan.

Session 7: Database Visualization

Value of Database Visualization

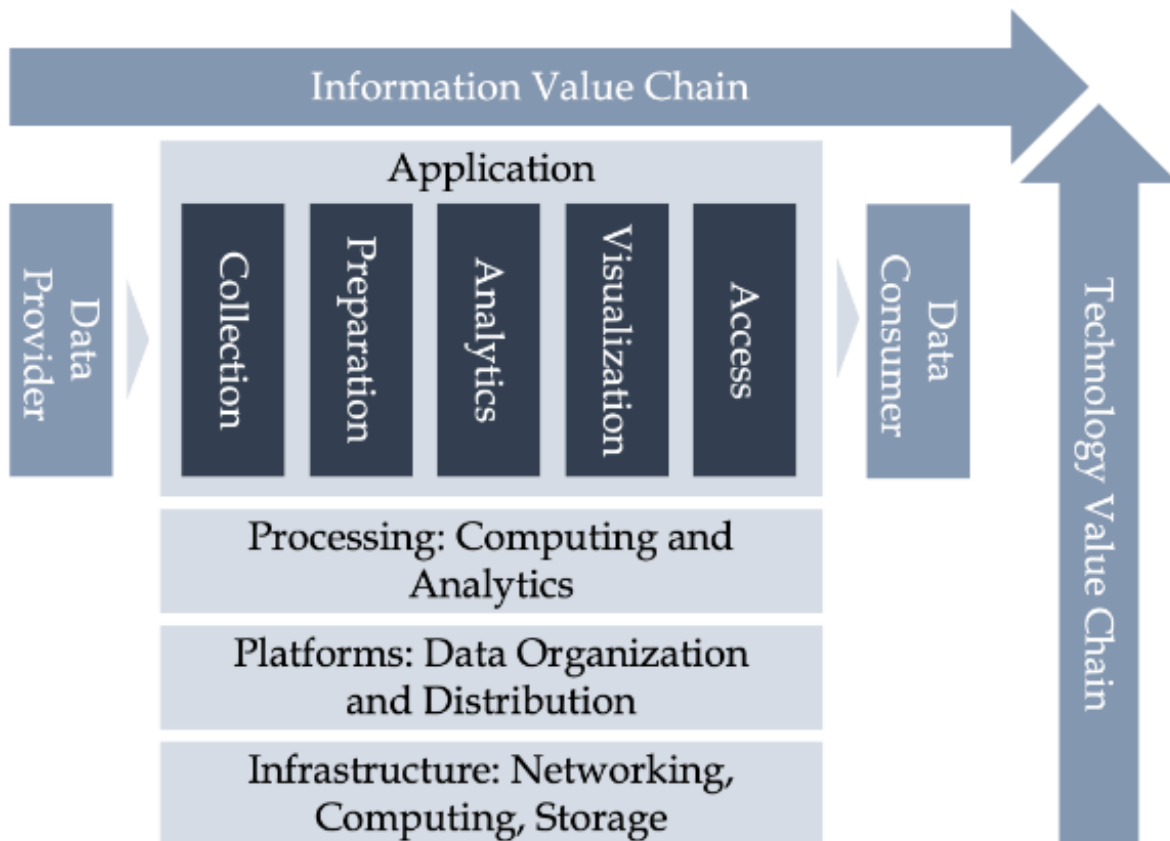


Figure 13: Value of Database Visualization